

Termination Analysis of Probabilistic Programs

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KLSS Seminar, Nov 26, 2025

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Termination of Turing Machines

The halting problem

To determine whether a given program (or TM) halts for a **given** input.

The halting problem is **undecidable** (Turing, 1937).

The universal halting problem

To determine whether a given program halts for **all** possible inputs.

Strictly more difficult than the halting problem.

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Collatz Conjecture

The “simplest” unsolved math problem:
to check if the program terminates for all $n \in \mathbb{N}$

```
while (n > 1){  
    if (n % 2 == 0)  
        n := n/2  
    else  
        n := 3*n+1  
}
```

How to prove termination?

Use a **variant function** on the program's state space whose value — on each loop iteration — is *monotonically decreasing* with respect to a (strict) well-founded relation.

Proof rules in Hoare logic

$$\frac{\{I \wedge G\}C\{I\}}{\{I\}\text{while } G \text{ do } C\{I \wedge \neg G\}} \quad \text{(partial correctness)}$$

$$\frac{\{I \wedge G \wedge V = z\}C\{I \wedge V < z\}}{\{I\}\text{while } G \text{ do } C\{I \wedge \neg G\}} \quad \text{(total correctness)}$$

- $I : \Sigma \rightarrow \{\top, \perp\}$ is the **invariant** — what is not changing.
- $V : \Sigma \rightarrow \mathbb{N}$ is the **variant** — what is changing
 - also called the ranking function.

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Probabilistic Programs

What?

- Programs with **random assignments**, **conditioning** and usual control-flow constructs

Why?

- for describing randomized algorithms.
- for designing probabilistic models.

Syntax and Semantics

- Probabilistic Guarded Command Language (pGCL) syntax:

$$C \hat{=} \text{skip} \mid x \leftarrow e \mid x \sim \mu \mid C_1; C_2 \mid C_1[p]C_2 \mid \\ \text{if}(\varphi) C_1 \text{ else } C_2 \mid \text{while}(\varphi) C$$

- $x \sim \mu$: the value of x is sampled from a distribution μ .
- $C_1[p]C_2$: choose C_1 with prob p and choose C_2 with prob $(1 - p)$.
- Remember: a program state is a **distribution** (rather than a value)

Termination Analysis

For deterministic programs

- A program either terminates or not on a given input.
- Terminating programs have a finite run-time.
- The property “having a finite run-time” is compositional.

All these facts do not hold for probabilistic programs!

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Certain Termination

```
while (x > 0) {  
    x := x-1 [1/2]  x := x-2  
}
```

This program **terminates** for all integer inputs x .

Almost-Sure Termination (AST)

```

x = 1;
while (x > 0) {
    x := 0 [1/2]  x := 1
}

```

This program does not always terminate.

It **diverges** with probability 0.

It **almost surely terminates** (with probability 1)
in finite expected time.

$$\mathbb{E}[T] = \sum_{i=1}^{\infty} i \cdot \mathbb{P}(T = i) = \sum_{i=1}^{\infty} i \cdot \left(\frac{1}{2}\right)^{i-1} \cdot \left(1 - \frac{1}{2}\right) = 2$$

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AST with infinite expected runtime

```
x = 1;  
while (x > 0) {  
    x := x-1 [1/2] x := x+1  
}
```

1D symmetric random walk almost surely terminates,
but requires an infinite expected time.

Not Almost-Sure Termination

```
function P:  
  skip [1/2] {call P; call P; call P}
```

This program terminates with probability $\frac{\sqrt{5}-1}{2}$.

$$x = \frac{1}{2} + \frac{1}{2}x^3$$

Problems

- **Qualitative analysis:** check almost surely termination (AST)
 - Positive AST: ... and in an expected **finite** number of steps
 - “Unbounded” AST: ... and in an expected **infinite** number of steps
- **Quantitative analysis:** to compute a lower/upper bound $p \in [0, 1]$ on termination probability.

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Martingales

- A stochastic process $X = \{X_0, X_1, X_2, \dots\}$ is a **martingale** if
 - $\{X\}_{n \in \mathbb{N}}$ is adapted to a filtration $\{\mathcal{F}\}_{n \in \mathbb{N}}$;
 - X_n is integrable for each n , i.e., $\mathbb{E}(|X_n|) < \infty$,
 - with probability 1 (i.e., almost surely),

$$\mathbb{E}[X_{n+1} \mid \underbrace{X_0, X_1, \dots, X_n}_{\mathcal{F}_n}] = X_n.$$

- ... is a **supermartingale** if

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n.$$

Ranking Supermartingales

A discrete-time stochastic process $X = \{X_n\}_{n \in \mathbb{N}}$ adapted to a filtration $\{\mathcal{F}_n\}_{n \in \mathbb{N}}$ is a **ranking supermartingale (RSM)**

- (integrability) $\mathbb{E}[|X_n|] < \infty$ for $n \in \mathbb{N}$;
- (lower-bound) $\exists k \leq 0$ s.t. $X_n \geq k$ almost surely for $n \in \mathbb{N}$;
- (decrease) $\exists \epsilon > 0$ s.t. $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n - \epsilon \cdot \mathbf{1}_{X_n \geq 0}$ almost surely.

ranking function: **decrease** by ϵ at each step.

RSM: **expected** to **decrease** by ϵ at each step.

RSM for AST

A RSM with a positive initial position
almost surely becomes ≤ 0 in **finite** number steps

$\mathbb{P}(Z_X < \infty) = 1$ and $\mathbb{E}[Z_X] \leq \frac{\mathbb{E}[X_0]}{\epsilon}$
where Z_X denotes the first time X drops below 0.

RSM Maps

Use a function V to **map** program states to a stochastic process?

- An **invariant** $\Omega \subseteq \mathbb{R}^n$ as an over-approximation of the reachable set.
- A **variant** function $V : \Omega \rightarrow \mathbb{R}$

$$V(x) \leq 0 \qquad \forall x \in T \text{ (terminal states)}$$

$$0 < V(x) \leq H \qquad \forall x \in \Omega \setminus T$$

$$\mathbb{E}[V(x')] \leq V(x) - \epsilon \qquad \forall x \in \Omega \setminus T$$

where x' is the value of x after one loop.

RSM Maps

Theoretical guarantees?

- **Sound**

an RSM exists $\implies$ PAST

- **Complete** for $\mathbb{Z}$ -valued programs

PAST $\implies$ an RSM exists

Synthesis?

- **Template-based**: Assume $V(x)$ is of certain parametric forms, usually a **linear** or **polynomial** expression.

RSM Maps

Theoretical guarantees?

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AST with infinite expected runtime

```
while ( 0 < x ) {  
    x := x-1 [1/2] x := x+1  
}
```

1D random walk still beyond reach.

“Walled” programs

```
while ( 0 < x < M ) {  
    x := x-1 [1/2] x := x+1  
}
```

After adding a **wall**, the program becomes AST.

Borel–Cantelli Lemma (0-1 Law)

- Let $E_1, E_2, \dots$ be a sequence of events in some probability space.
- First Borel-Cantelli Lemma**

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty \implies \underbrace{\mathbb{P}(\limsup_{n \rightarrow \infty} E_n) = 0}_{\text{a.s. } E_n \text{ occur finitely}}$$

- Second Borel-Cantelli Lemma.** If the events E_n are **independent**

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) = \infty \implies \underbrace{\mathbb{P}(\limsup_{n \rightarrow \infty} E_n) = 1}_{\text{a.s. } E_n \text{ occur infinitely}}$$

Example: the infinite monkey theorem.

Parametric RSM

A non-negative stochastic process $X = \{X_n\}_{n \in \mathbb{N}}$ is a **parametric ranking supermartingale (PRSM)** if

- for each $n \in \mathbb{N}$, $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n$;
- there exist functions $d : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ and $p : \mathbb{R} \rightarrow [0, 1]$, both **non-increasing and strictly positive on $\mathbb{R}_{>0}$** s.t. for each $n \in \mathbb{N}$:

$$\mathbb{P}(X_{n+1} \leq X_n - d(X_n)) \geq p(X_n).$$

there is a **positive** probability of a **strict** decrease.

AST with infinite expected runtime

```

while ( 0 < x ) {
    x := x-1 [1/2] x := x+1
}

```

Let $V(x) := x$, $d(x) := 1$, $p(x) = 1/2$.

$$\mathbb{E}[x] \leq x$$

$$\mathbb{P}(x' \leq x - 1) \geq \frac{1}{2}$$

there is a positive probability ($\frac{1}{2}$) of a strict decrease (by 1)

Completeness

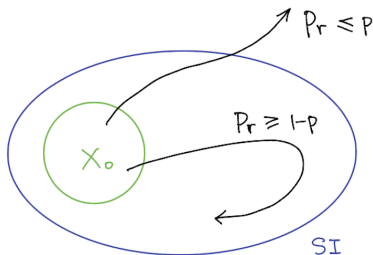
- PRSM rule is proposed for pWhile loops, conjectured to be incomplete (McIver, POPL'18).
- (Majumdar, L-CSS'24): **complete** for discrete-time **weak Feller process**
 - The transition semigroup maps bounded continuous functions to bounded continuous functions.
- (Majumdar, POPL'25): **complete** for probabilistic control flow graphs.
 - restrictions: only discrete distribution, variables take rational values ...
 - another Drift-and-Variant rule.

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Probabilistic Termination

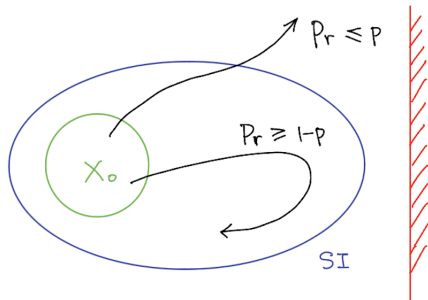
- Probabilistic Termination: to find **lower/upper bound** for termination.
- A **p -stochastic invariant (p -SI)** is a set Ω such that

$$\mathbb{P}(\text{a trajectory starting from } X_0 \text{ leaves } \Omega) \leq p$$



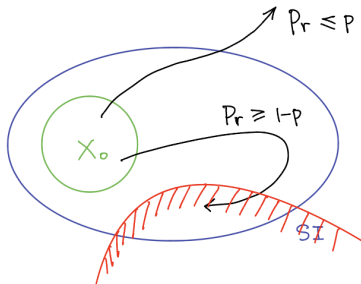
Probabilistic Termination Upper Bound p

Find a p -SI Ω s.t. $\Omega \cap X_T = \emptyset$.



Probabilistic Termination Lower Bound $1 - p$

Find a p -SI Ω
and prove almost surely reach $\Omega \cap X_T$ within Ω .



Summary:

- Qualitative: $AST = \text{positive AST} + \text{unbounded AST}$
 - positive AST: **Ranking Supermartingales (RSM)**.
 - unbounded AST: **Parametric RSM**.
- Quantitative: find upper/lower bounds.
 - reduced to AST by **Stochastic Invariants**.

Thanks! & Questions?

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